The Distinguished Affine Root at the Boundary: An Adversarial Audit and a Correct Valuative Theorem

Repository audit draft

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Abstract

We audit the chain H1–H3 concerning the Hurwitz compactification of rerooting, stacky extension of a selected zero-fiber root, and descent of that root to coarse moduli. We distinguish three assertions that the current repository conflates: the elementary valuative extension of a rational point through a finite cover; the quotient-stack description of choosing one element of an unordered divisor; and the comparison of that quotient stack with the repository-specific admissible-cover contraction and regular-reconstruction component. The first two assertions are proved here, including arbitrary simultaneous collisions and stabilizers. The third is isolated as an explicit comparison hypothesis. Under that hypothesis we give a complete proof that the distinguished affine-root marking extends uniquely over every DVR, as a scheme-valued marking and not merely as a geometric point. We then show why the present H1–H3 text does not prove the comparison hypothesis. In particular, properness of an ambient admissible-cover stack, equality of generic function algebras, and isomorphisms of completed collision charts do not by themselves construct the required global finite marked cover. The resulting verdict is that the valuative statement is correct conditionally and locally well supported, but H2 and H3 are not presently available as foundational infrastructure.

Contents

1 Verdict and logical shape

Fix a field k of characteristic zero and an integer $N \geq 4$, and put $n = N - 2$. On the clean seed locus a normalized polynomial

$$H(W) = -\frac{W^2(W-1)\prod_{i=1}^{n-1}(W-\rho_i)}{\prod_{i=1}^{n-1}(1-\rho_i)}$$

has an exact double zero at 0, a distinguished simple zero at 1, and $n-1$ further simple nonzero zeros. The open rerooting cover forgets which of the n simple zeros was normalized to 1. The repository asks whether the root selected by the regular-reconstruction chart continues uniquely when some or all of these roots collide.

There is a short correct answer once the right finite morphism has actually been constructed. If R is a DVR with fraction field K, if $Y \to X$ is finite, and if an R-point of X has a K-rational lift to Y, that lift extends uniquely to R whenever R is integrally closed. This is the entire scheme-valued valuative argument. It works for a nonreduced collision fiber and does not infer uniqueness from the number of geometric points.

The hard point is therefore upstream:

> construct a global finite marked object over the precise compactification, and identify its generic section with the distinguished regular-reconstruction component.

The current H2/H3 proof assumes this in the phrases "the contracted ambient root cover" and "the closure of the distinguished component." H1 cites proper admissible-cover technology but does not define the repository subfunctor, construct its contraction, or prove that it is the quotient compactification used in the later invariant-ring calculation.

Our conclusion is consequently two-part.

1. The quotient-stack and coarse collision model has the desired DVR extension theorem. We prove it in Sections 2–5.

2. Application to the claimed Hurwitz–LL compactification is conditional on the global comparison package in Definition 6.1. The repository does not yet prove that package; Section 7 records the precise gaps.

This distinction matters for OP-LR-NE. Unique specialization of one point of a finite cover is not properness of a relative Isom functor. Even after the comparison package is proved, OP-LR-NE still requires a bounded, representable, closed extension of isomorphisms and a proof that its projection is proper and quasi-finite.

2 The moduli functors

We define all objects used in the valuative statement. The definitions are deliberately independent of any chosen presentation by coefficients.

2.1 Stable root configurations

Let $L = \{0, \infty, 1, \ldots, n\}$, a set of $n+2 = N$ labels. Write $\overline{M}_{0,L}$ for the fine moduli scheme of stable genus-zero curves with pairwise disjoint sections $(p_\ell)_{\ell\in L}$. Thus an object over a scheme T is a proper flat nodal curve $C \to T$ of arithmetic genus zero, together with sections through the smooth locus, such that every geometric component has at least three special points. The symmetric group S_n permutes the labels $1, \ldots, n$ and fixes $0, \infty$.

For a finite constant group G acting on a scheme U, we use the following functorial definition of the quotient stack. An object of $[U/G](T)$ is a right G-torsor $P \to T$ in the etale topology together with a G-equivariant morphism $P \to U$. A morphism is an isomorphism of torsors compatible with the maps to U. Set

$$\mathscr{B}_n = [\overline{M}_{0,L}/S_n], \qquad \mathscr{S}_n = [\overline{M}_{0,L}/S_{n-1}], \tag{2.1}$$

where $S_{n-1} \subset S_n$ fixes the label 1. We call $\mathscr{B}_n$ the unmarked collision-separating root stack and $\mathscr{S}_n$ the marked one. The morphism

$$q : \mathscr{S}_n \longrightarrow \mathscr{B}_n \tag{2.2}$$

forgets which simple-root label is 1.

The word "marked" in (2.1) does not mean that every label is globally ordered on the base. It means that one element of the unordered degree-n divisor is selected; the other $n-1$ elements remain unordered. Stable bubbles keep all labelled points disjoint even when their contraction has a multiple root.

2.2 The open normalized-seed functors

Let $\mathscr{U}_n \subset \mathscr{B}_n$ be the open on which the source curve is smooth and the points $p_0, p_\infty, p_1, \ldots, p_n$ are distinct. On an etale cover where the n simple points are labelled, the unique coordinate W satisfying $p_0 = 0$ and $p_\infty = \infty$ is still subject to scaling. Choosing one simple point as $p_1 = 1$ removes that scaling. The corresponding polynomial is, up to the endpoint derivative normalization,

$$W^2 \prod_{i=1}^{n} (W - p_i).$$

We denote the marked open by $\mathscr{U}_n^{\mathrm{mark}} = q^{-1}(\mathscr{U}_n)$. The repository's clean normalized seed space maps to $\mathscr{U}_n^{\mathrm{mark}}$ by sending H to its zero divisor with 0, ∞, and the root 1 distinguished.

This map is elementary on the open. Nothing in its formula constructs an extension to a boundary compactification: when points collide, a stable configuration acquires bubbles, while a coefficient model contracts them. Relating those two operations is a separate theorem.

2.3 Admissible polynomial covers

We next give a sufficiently precise functor for the external compactification technology. Fix the ramification profile consisting of total ramification over a target section b_∞ and profile $(2, 1^n)$ over a target section b_0. Let $\mathscr{H}_N^{\mathrm{adm}}(T)$ be the groupoid of data

$$(C \xrightarrow{f} P; b_0, b_\infty; p_0, p_\infty; D), \tag{2.3}$$

where $P \to T$ is a stable pointed nodal genus-zero target (possibly twisted at nodes and markings), $C \to T$ is a nodal source, f is a finite flat degree-N admissible cover, p_∞ is the totally ramified point over b_∞, p_0 has ramification index two over b_0, and D is the residual degree-n unramified divisor in $f^{-1}(b_0)$. At nodes the two branch indices agree, and the usual stability condition makes automorphism groups finite. Branch data away from b_0, b_∞ must also be fixed in a given connected Hurwitz component; omitting it does not define a finite-type Hurwitz stack.

The marked functor $\mathscr{H}_N^{\mathrm{adm,mark}}$ adds a section $p_1 : T \to D$. In the collision-separating model D is represented by disjoint markings on the expanded source. Forgetting p_1 gives

$$q_{\mathrm{adm}} : \mathscr{H}_N^{\mathrm{adm,mark}} \longrightarrow \mathscr{H}_N^{\mathrm{adm}}. \tag{2.4}$$

To obtain a compactification of the seed locus one must specify: the Hurwitz component, all branch markings or their allowed collisions, the target stability weights, and the closure of the normalized pencil locus. ACV and Deopurkar provide ambient stacks after these choices; their properness does not choose the repository-specific component.

2.4 The contracted root incidence

For a monic degree-n polynomial

$$F(T) = T^n - e_1 T^{n-1} + \cdots + (-1)^n e_n,$$

the universal root incidence is

$$\mathcal{R}_n = \operatorname{Spec} k[e_1, \ldots, e_n, T]/(F(T)) \longrightarrow \mathbb{A}^n_{e_1,\ldots,e_n}. \tag{2.5}$$

It is finite flat of degree n. It is not etale along the discriminant. For a family of zero divisors, the contracted root cover is obtained by base change from (2.5), with the fixed double root removed before forming the degree-n residual polynomial. A marking is a section of this finite cover, equivalently a factorization $F = (T - a)G$ over the base.

The inverse-pencil incidence

$$X_H = \operatorname{Spec} k[s, t, W]/(H(W) - sW + t) \longrightarrow \mathbb{A}^2_{s,t} \tag{2.6}$$

is instead finite flat of degree N. It parametrizes every point in the fiber of $H(W) - sW$ over $-t$. It must not be identified without argument with the degree-n cover (2.5), which only selects residual simple roots of the zero fiber after removing the fixed double root.

3 The quotient-stack theorem

Proposition 3.1. *The morphism q in (2.2) is representable, finite, and etale of constant degree n. For every geometric point $x : \operatorname{Spec}\Omega \to \mathscr{B}_n$, its base change is the discrete n-element scheme of choices of one label.*

Proof. Let $X = \overline{M}_{0,L}$, $G = S_n$, and $H = S_{n-1}$. Given a scheme $T \to [X/G]$, let $P \to T$ be the associated G-torsor. A reduction of its structure group to H is a section of the associated bundle

$$P \times^G (G/H) \longrightarrow T. \tag{3.1}$$

The fiber product $T \times_{[X/G]} [X/H]$ is canonically this associated bundle, with the compatibility map to X carried along. Since G/H is a constant etale scheme with n elements, (3.1) is finite etale of degree n. The base change by every scheme is a scheme, proving representability. □

Remark 3.2. *Proposition 3.1 is not contradicted by ramification on coarse spaces. The stack map remembers stabilizers. The coarse map forgets them and is ramified at collisions. Conflating these two maps is one of the main hazards in H2/H3.*

Proposition 3.3 (relative automorphisms)**.** *Let y be an object of $\mathscr{S}_n$ with image x in $\mathscr{B}_n$. Then*

$$\operatorname{Aut}_{\mathscr{S}_n}(y) \longrightarrow \operatorname{Aut}_{\mathscr{B}_n}(x)$$

is injective. Hence an automorphism of a marked lift inducing the identity on the unmarked object is the identity.

Proof. This is the inertia criterion for representability, or directly the torsor description: an automorphism of an H-reduction that is the identity on the underlying G-torsor is the identity. Equivalently, the relative inertia of q is trivial. □

At a collision, the unmarked object can still have nontrivial absolute stabilizer. For example, a symmetric cluster of μ labels can carry an S_μ stabilizer, while a choice of one label reduces it to $S_{\mu-1}$. Proposition 3.3 says that this does not create a hidden automorphism of the lift over the fixed unmarked object. It does not say that the coarse space remembers the labelled point.

4 Coarse root charts and simultaneous collisions

We now prove the invariant-ring calculation used by H3 and state exactly what it does and does not imply.

Proposition 4.1 (one cluster)**.** *Let $A = k[x_1, \dots, x_\mu]$, with S_μ permuting the variables and $S_{\mu-1}$ fixing x_1. If e_j denotes the elementary symmetric function in all variables, then*

$$A^{S_{\mu-1}} \cong A^{S_\mu}[T]/(T^\mu - e_1 T^{\mu-1} + \cdots + (-1)^\mu e_\mu), \tag{4.1}$$

with $T \mapsto x_1$. In particular the coarse forgetful map is finite flat of degree μ, is generically etale, and has total-collision fiber $k[T]/(T^\mu)$.

Proof. Put $e'_j = e_j(x_2, \ldots, x_\mu)$. Then $e_j = e'_j + Te'_{j-1}$, with $e'_0 = 1$. Recursively, every e'_j belongs to $k[e_1, \ldots, e_\mu, T]$. Hence $A^{S_{\mu-1}} = k[T, e'_1, \ldots, e'_{\mu-1}]$ is generated over $A^{S_\mu} = k[e_1, \ldots, e_\mu]$ by T. Conversely the formulas $e_j = e'_j + Te'_{j-1}$ define the inverse change of generators. The last recursion equation $e_\mu = Te'_{\mu-1}$ becomes exactly the displayed monic equation after eliminating $e'_1, \ldots, e'_{\mu-1}$. Thus the two presentations are inverse isomorphisms. Since the equation is monic, the map is finite flat of rank μ. Setting every e_j to zero gives the stated fiber. $\square$

Warning 4.2. *The special fiber in Proposition 4.1 has one geometric point but μ scheme-theoretic length. The fact that its topological space has one point does not prove uniqueness of a section over a DVR. Uniqueness is instead a consequence of separatedness after the generic section has been fixed; existence uses integrality, as in Section 5.*

Proposition 4.3 (simultaneous multicluster chart)**.** *Suppose the special divisor has disjoint clusters of sizes $\mu_1, \ldots, \mu_r$, together with simple spectator roots. After an etale localization separating the cluster centers, the completed coarse root cover is a disjoint union, over the possible cluster containing the selected root, of completed tensor products of the corresponding rings (4.1) with formal power-series rings in spectator and center parameters. Its fiber over the deepest collision stratum has one geometric selected point for each distinct cluster, and length μ_i on the component selecting cluster i.*

Proof. Choose pairwise disjoint etale neighborhoods of the distinct cluster centers. Hensel factorization writes the residual monic polynomial uniquely as a product $F_1 \cdots F_r F_{\mathrm{sp}}$, where F_i has degree μ_i and F_{sp} is etale. The functor of choosing a root is the disjoint union of the functors of choosing a root of one factor. Completion commutes with finite products and with the finite base changes in (4.1). The parameters of the other factors are formally independent, giving the asserted completed tensor products. Specializing each F_i to T^{μ_i} gives the length and geometric-point statements. $\square$

The proposition corrects an imprecision in the phrase "simultaneous clusters have one geometric coarse mark." There is one point within a chosen cluster, but different cluster centers remain different geometric points. A generic distinguished root determines which component is chosen; the collision alone does not.

5 The finite-cover DVR lemma

We now prove the scheme-valued extension result in the generality actually needed.

Lemma 5.1 (finite valuative extension)**.** *Let R be a DVR with fraction field K, let X be an algebraic space, and let $f : Y \to X$ be finite and representable. Given a commutative generic diagram*

$$\begin{array}{ccc} \operatorname{Spec} K & \longrightarrow & Y \\ \downarrow & & \downarrow f \\ \operatorname{Spec} R & \longrightarrow & X, \end{array} \tag{5.1}$$

there is a unique dotted arrow $\operatorname{Spec} R \to Y$ *completing the diagram. The same assertion holds after arbitrary extension of DVRs.*

Proof. Base change by $\operatorname{Spec} R \to X$. Because f is representable and finite, the fiber product is a finite algebraic space over $\operatorname{Spec} R$, hence an affine scheme $\operatorname{Spec} B$ with B finite over R. The generic lift is an R-algebra map $B \to K$. Every element of B is integral over R, so its image in K is integral over R. A DVR is integrally closed in K; therefore the image lies in R. This gives an R-algebra map $B \to R$ and hence the extension. Since a finite morphism is separated, two extensions agree if they agree on the dense generic point. The proof applies verbatim to any DVR extension. $\square$

Remark 5.2. *Etaleness is unnecessary in Lemma 5.1. This is important for the coarse map, which is ramified at collisions. Properness alone would give existence only in the usual valuative sense and may require attention to field extensions for stacks. Finiteness plus a K-rational lift gives the stronger extension over the original DVR.*

Corollary 5.3 (stacky marking)**.** *For every DVR map $\operatorname{Spec} R \to \mathscr{B}_n$, every lift of its generic point to $\mathscr{S}_n$ extends uniquely, up to a unique 2-isomorphism, to $\operatorname{Spec} R \to \mathscr{S}_n$. There is no relative stabilizer ambiguity.*

Proof. Apply Proposition 3.1 and Lemma 5.1. Representability turns the lifting problem into one for a finite scheme. Proposition 3.3 makes the relative 2-automorphism group trivial. $\square$

Corollary 5.4 (coarse marking)**.** *Let B_n and S_n be the coarse spaces of $\mathscr{B}_n$ and $\mathscr{S}_n$, and suppose the induced map $S_n \to B_n$ is finite. Then a K-rational generic marked lift of an R-point of B_n extends uniquely to an R-point of S_n. In the local collision charts this extension is the unique integral root of the universal monic polynomial that equals the given generic root.*

Proof. Apply Lemma 5.1 to $S_n \to B_n$. In chart (4.1), the generic marking is a map sending T to an element $a \in K$ integral over R. Thus $a \in R$, and the factorization by $T - a$ extends. This proves existence as a scheme-valued section. Separatedness proves uniqueness. $\square$

Example 5.5. *The family $F(T) = T^2 - \pi^2$ over a DVR has two generic roots $\pm\pi$ and both specialize to the unique geometric point of $k[T]/(T^2)$. Once the generic root $+\pi$ is fixed, only the section $T \mapsto \pi$ extends it. This shows simultaneously why one geometric special point is insufficient for scheme-valued uniqueness and why the finite-cover lemma gives the right answer.*

6 The comparison package actually required

The preceding results apply to the repository only after the following data are constructed. We state them as a package so that no conclusion is hidden inside notation.

Definition 6.1 (global comparison package)**.** *A Hurwitz compactification of the normalized seed locus satisfies* (CMP) *if it comes with:*

1. *a proper normal Deligne–Mumford stack $\overline{\mathscr{B}}_N^{\mathrm{seed}}$ containing the unmarked clean seed stack, and a proper normal marked stack $\overline{\mathscr{S}}_N^{\mathrm{seed}}$ containing the distinguished marked seed stack;*
2. *a representable finite morphism $q_{\mathrm{seed}} : \overline{\mathscr{S}}_N^{\mathrm{seed}} \to \overline{\mathscr{B}}_N^{\mathrm{seed}}$ restricting to the degree-n rerooting cover;*
3. *a global contraction to a finite degree-n residual-root cover $\mathcal{R}_N^{\mathrm{seed}}$ and a morphism $\overline{\mathscr{S}}_N^{\mathrm{seed}} \to \mathcal{R}_N^{\mathrm{seed}}$ whose open restriction is the selected nonzero zero-fiber root;*
4. *an identification of the open section in item 3 with the unique regular-reconstruction component supplied by F2;*

5. *compatibility of items 1–4 with coarse spaces and base change, including a finite coarse marked map and the invariant charts of Propositions 4.1 and 4.3;*

6. *if normalized Stein factors or conductor squares are to be used later, a constructed global comparison morphism identifying those finite models, not only their generic fields or completed local rings.*

Theorem 6.2 (distinguished affine-root extension, conditional form)**.** *Assume* (CMP)*. Let R be any DVR with fraction field K, and let $\operatorname{Spec} R \to \overline{\mathscr{B}}_N^{\text{seed}}$ be a degeneration whose generic point lies in the clean locus. Suppose the distinguished affine root gives a K-lift to $\overline{\mathscr{S}}_N^{\text{seed}}$. Then:*

1. *the lift extends over R;*

2. *it is unique as a scheme-valued lift, up to a unique 2-isomorphism;*

3. *its image on coarse moduli is the unique extension of the coarse generic mark;*

4. *the assertion commutes with arbitrary DVR base change;*

5. *at simultaneous clusters of sizes $\mu_1, \dots, \mu_r$, the extension lies in the component determined by the generic root and has special scheme-theoretic length μ_i if that root enters cluster i;*

6. *no automorphism acting trivially on the unmarked degeneration can change the extended mark.*

Proof. Item 1 follows from Lemma 5.1 applied to q_{seed}. Item 2 follows from separatedness and representability; the latter kills relative inertia exactly as in Proposition 3.3. By item 5 of (CMP), the induced coarse marked map is finite, so Corollary 5.4 gives item 3. Both the integral-closure proof and the uniqueness proof survive DVR extension, giving item 4.

For item 5, pass etale-locally to the Hensel-separated factorization used in Proposition 4.3. The generic root selects exactly one factor. Its closure cannot jump to another open-and-closed factor, and the chosen factor has collision algebra $k[T]/(T^{\mu_i})$. This proves compatibility with all simultaneous clusters, including mixed pair, triple, and higher collisions. Finally, item 6 is the injectivity on stabilizers implied by representability. Item 4 of (CMP) identifies this abstractly extended mark with the distinguished regular-reconstruction component rather than an arbitrary root label. □

Corollary 6.3. *Under* (CMP)*, the closure of the graph of the distinguished section inside the finite marked cover is isomorphic to the normal DVR base along every trait. If the same graph is globally finite and birational over a normal integral marked coarse compactification, then it is globally isomorphic to that compactification.*

Proof. The trait statement is Theorem 6.2. The global statement is the finite-birational criterion. Notice that global finiteness and the existence of the morphism are hypotheses; they do not follow from a completed-local description alone. □

7 Adversarial audit of H1–H3

We compare the current manuscript with the hypotheses used above. The point is not that the desired comparison is false. It is that several nontrivial morphisms are invoked before they are defined.

7.1 H1: the compactification is not specified enough

The current proof says that taking the normalization of the closure in "the proper admissible-cover stack" gives the desired compactification. There is no single such stack. One must fix the Hurwitz component, all ramification profiles, branch markings, allowed branch collisions, target weights, and rigidifications. Deopurkar's construction explicitly varies with the compactification of the branch divisor. Ambient properness proves properness of a closed substack after it has been defined; it does not prove that the closure equals $[\overline{M}_{0,N}/S_n]$ or that a coefficient contraction exists.

The displayed normalized polynomial is determined by its zero divisor on the smooth locus. At the boundary, however, a stable pointed source and an admissible map carry node indices, target expansions, and possibly twisted structure not encoded by the stabilized zero divisor alone. A proof of H1 must construct mutually inverse functors, or at least a finite birational morphism with a verified normal target, between the relevant closure and the claimed quotient stack. The manuscript currently passes from one to the other with the word "equivalently."

The LL assertion introduces an additional mismatch. A polynomial cover is compactified by branch data, while $\overline{M}_{0,N}$ in H1 records zero-fiber source points. These are related by the polynomial, but their stable limits need not be identical compactifications. The stable branch-divisor theorem extends the branch divisor of a family; it does not identify the closure of the zero-root configuration with a prescribed Hurwitz component.

7.2 H2: tautological persistence versus comparison

On a marked admissible-cover stack, the marked section persists by definition. This proves no descent statement and no identification with the repository reconstruction cover. The quotient calculation in Proposition 3.1 is correct if the marked and unmarked stacks really are the two quotient stacks in (2.1). That identification is the missing H1 comparison, not a consequence of subgroup index.

The formal-comparison proof uses two finite normal algebras with the same generic function algebra and invokes uniqueness of integral closure. This argument is valid only after proving all of the following: the proper incidence has a coherent Stein algebra finite over the stated base; its generic algebra is the full algebra of (2.6), with the correct product of components; the normalization is taken in that algebra; and a global comparison map to the repository incidence has been constructed. None of these is supplied by the phrase "pull back and mark a point over $-t$."

Moreover, that phrase defines the degree-N incidence (2.6), whereas H2's rerooting morphism has degree $n = N - 2$. Removing the fixed length-two zero cluster and restricting to the zero fiber are operations that must be made functorial in a degeneration. Equality of generic fields cannot silently change the degree.

The completed chart $k[[s, u]]$ is a correct chart for the smooth hypersurface X_H. An isomorphism of this chart with a contracted admissible chart is asserted, not derived from an explicit contraction. Even isomorphisms at every completion would not automatically glue to a global isomorphism without a global morphism and suitable finiteness. The conductor calculation similarly compares two conductor squares only after the downstairs finite subalgebras have been identified; equality of normalizations and numerical conductor exponents is insufficient.

7.3 H3: the finite-birational argument starts after the gap

The finite-birational criterion used by H3 is correct. Its application is circular. The proof places a closure Z in "the contracted finite root cover," declares it finite over the marked coarse compactification, and then uses normality. A closure in a finite ambient space is finite only when that ambient space and its map over the same base have already been constructed. This is item 3 of (CMP), the main missing comparison.

There is also a base mismatch in the statement. The universal root incidence $S_n \to B_n$ is the coarse marked-to-unmarked map. A graph of the tautological selected root over S_n lives in the pullback of the root cover to S_n. Saying that its closure is finite birational over S_n is useful only after constructing this graph and proving that the generic regular-reconstruction component maps to it. The relative Hilbert scheme of length-one subschemes of a finite scheme is indeed the finite scheme itself, but this identity does not choose the desired component or prove the comparison with the admissible contraction.

Finally, the categorical property of a coarse moduli space only factors maps from the stack to algebraic spaces. It does not turn a stack section of a universal curve into a section of a coarse universal curve when no such universal curve exists, nor does it prove independence of distinct stacky lifts of the same coarse trait. Here independence can be proved from the explicit finite coarse root incidence and Lemma 5.1, but the coarse map must first be identified globally.

7.4 Risk matrix

Claim	What is valid	What is not yet proved
Marked/unmarked quotient	$[X/S_{n-1}] \to [X/S_n]$ is representable finite etale of degree n	The seed admissible-cover closures are these quotient stacks
Coarse collision chart	The invariant ring is the universal monic-root incidence and is ramified	This chart is the coarse contraction of the repository's global Hurwitz closure
Stein comparison	A finite normal model is determined by a fixed generic algebra	The proper incidence has that exact generic algebra and admits the claimed global comparison map
Completed charts	Hensel factorization separates distinct clusters	The local identifications arise from and glue to one global contraction
Conductor	Formulae can be checked once the finite downstairs algebra is fixed	Equal exponents alone identify the conductor square
DVR uniqueness	A fixed generic lift through a finite map extends uniquely over a DVR	One geometric point implies a unique scheme-valued lift, or an ambient proper stack makes a specific Isom relation proper
Stabilizers	Representability kills relative automorphisms	Coarse descent by itself remembers all stacky choices

8 Consequences for OP-LR-NE

Theorem 6.2, once unconditional, supplies exactly one ingredient for OP-LR-NE: a chosen affine sheet cannot jump between components of the finite marked root cover along a DVR. It does not establish valuative properness of an isomorphism relation.

Let $\mathcal{X}$ and $\mathcal{Y}$ be two families of marked normalized incidence covers over a base T. A candidate functor

$$\mathcal{I} = \mathrm{Isom}_T^{\mathrm{dec}}(\mathcal{X}, \mathcal{Y}) \tag{8.1}$$

must specify which decorations are preserved: the finite cover, Hessian divisor, affine sheet, conductor, node pairing, and any target/source polarizations needed for boundedness. To infer that the closure of $\mathcal{I}$ is finite and proper, one still needs:

1. representability of (8.1), including control of automorphisms at stacky boundary points;
2. finite type or a boundedness theorem for the allowed isomorphisms;
3. separatedness, usually from a rigid enough decoration;
4. existence of extensions over DVRs, not only extension of the selected root;
5. quasi-finiteness at every boundary point, not merely generic deck rigidity.

Generic triviality of a deck group gives generic quasi-finiteness. It does not exclude new automorphisms on special fibers. Properness of the ambient Hurwitz stack does not imply properness of an open or locally closed Isom subfunctor. Thus OP-LR-NE remains genuinely open after the marking theorem, exactly as the repository's own final paragraph suggests.

9 A proof checklist that would close H1–H3

The following finite list turns the comparison package into verifiable lemmas.

1. Define the fully labelled seed Hurwitz functor, including every branch profile, target marking, stability weight, and rigidification.
2. Construct the morphism from the clean labelled zero-root configuration to that functor and prove it is an open immersion into one specified connected component.
3. Extend the construction over $\overline{M}_{0,N}$, or construct a proper birational morphism from the normalization of the Hurwitz closure to $\overline{M}_{0,N}$ and prove it is an isomorphism. Check node indices and twisted stabilizers on every boundary graph.
4. Prove equivariance for S_n and S_{n-1} before taking quotients. This yields the finite representable etale stack map without ambiguity.
5. Construct, over the unmarked closure, the residual zero-fiber divisor of degree n after removing the fixed ramification-length-two summand. Prove it is finite flat and commutes with base change.
6. Construct the contraction from the expanded source to that residual root cover. Verify that the selected section maps to its tautological root.
7. Identify the F2 regular-reconstruction component with this section on the clean open, then take its scheme-theoretic closure.
8. Prove the induced coarse map is finite and establish the local invariant presentation (4.1) from the global map. Do not reverse this implication.
9. For the degree-N inverse incidence and critical incidence, construct separate proper objects and global Stein comparison morphisms. Verify the generic algebras component by component before invoking integral closure.

10. Derive the conductor-square comparison from equality of the downstairs finite algebras; use completed exponent calculations as checks, not as the definition of the global isomorphism.

Once items 1–8 are complete, Theorem 6.2 gives the requested DVR extension immediately and without further admissible-cover technology. Items 9–10 are needed only for the stronger normalized-Stein and conductor claims bundled into H2.

10 Final theorem status

The requested sentence

> "The distinguished affine-root marking extends uniquely across every DVR degeneration in the claimed compactification"

has two readings. For the explicitly defined quotient compactification $\mathscr{S}_n \to \mathscr{B}_n$, it is proved by Corollary 5.3, Corollary 5.4, and Proposition 4.3. For the repository-specific Hurwitz/admissible-cover compactification and its regular-reconstruction contraction, it follows from Theorem 6.2 only after (CMP) is proved.

Accordingly, the adversarial status is:

- H1: partial; ambient compactification technology is available, but the seed component and quotient identification are not constructed.
- H2: partial; the abstract quotient-stack assertion is correct, while the normalized-Stein, contraction, and conductor comparisons are not proved globally.
- H3: plausible but high-risk; the local coarse chart and the finite-map DVR lemma are correct, but their application to the distinguished repository component depends on the missing global comparison.
- OP-LR-NE: open; it requires properness of a specific decorated Isom relation, which does not follow from H3 or from ambient properness.

This is not a counterexample to the intended theorem. It is a separation of the theorem's elementary valuative core from the global moduli comparison that must be supplied before the theorem can carry downstream dependencies.

A Explicit pair, triple, and mixed-cluster traits

This appendix checks the valuative mechanism in coordinates. These examples are not substitutes for the global comparison package, but they prevent the local statement from hiding a residue-field or nilpotent ambiguity.

A.1 A pair collision

Let $R = k[[\pi]]$ and consider

$$F(T) = (T - a - \pi)(T - a + \pi) = T^2 - 2aT + (a^2 - \pi^2).$$

The unmarked coefficient trait meets the discriminant at $\pi = 0$. The marked cover is

$$R[T]/((T-a)^2 - \pi^2).$$

It has two R-sections, $T = a + \pi$ and $T = a - \pi$, corresponding to the two generic roots. Both have the same special geometric value a. If the generic distinguished root is $a + \pi$, separatedness rules out the other section because they differ over K. Thus special geometric coincidence does not erase the generic label.

The stack picture contains more information. At the collision the unmarked configuration has a transposition exchanging the two colliding labels. Choosing one label replaces this stabilizer by the subgroup fixing that label, which is trivial inside the transposition group. The stack map remains etale: the ramification visible in the coarse equation is absorbed by this change of inertia. Consequently "finite etale" is correct for the stack map and false for its coarse counterpart.

A.2 A triple collision

Let

$$F(T) = (T - a - \pi)(T - a)(T - a + \pi).$$

Writing $U = T - a$ gives $F = U^3 - \pi^2 U$. The three generic markings extend as $U = \pi, 0, -\pi$. The total-collision fiber is $k[U]/(U^3)$, with one geometric point and length three. Again there are three different scheme-valued sections of the family, but no two extend the same generic section. At the symmetric special object the unmarked permutation group is S_3, while the stabilizer of a selected label is S_2; the relative inertia kernel remains trivial.

A cyclic transverse presentation such as $U^3 = \epsilon$ behaves differently over the unextended DVR $k[[\epsilon]]$: it has no generic K-rational root in general. After the ramified base change $\epsilon = \delta^3$ it acquires the roots $\delta, \zeta\delta, \zeta^2\delta$ when the cube roots of unity are present. This does not contradict Lemma 5.1. That lemma starts with a K-rational generic lift. Properness of a stack may produce a lift only after a finite base change, whereas H3's strong wording asserts an extension over the given DVR. The distinguished normalized root 1 is K-rational on the clean marked seed, so the stronger conclusion is available only after the marked finite cover has been globally identified.

A.3 Two simultaneous clusters

Let

$$F(T) = \prod_{i=1}^{\mu}(T - a - \alpha_i\pi) \prod_{j=1}^{\nu}(T - b - \beta_j\pi), \qquad a - b \in R^\times.$$

Hensel factorization separates the degree-μ and degree-ν factors over R. A generic root in the a-cluster cannot specialize into the b-cluster because the two factors define open-and-closed pieces of the finite root cover. Inside the a-piece all μ geometric roots can merge, leaving special algebra $k[U]/(U^\mu)$ after a total collision. The distinguished section still exists uniquely because its generic value is fixed. This proves the multicluster assertion without pretending that the entire special root cover has one geometric point: it has at least the two points a and b.

The same argument works when the cluster centers have nontrivial residue fields. After a finite separable extension of the residue field, Hensel factorization produces geometric factors permuted by Galois. A K-rational selected root defines a Galois-stable section, and the integral-closure argument descends it to R. If the root is defined only after a field extension, the conclusion is correspondingly a section only after normalization in that extension; one must not call it an R-valued mark.

B Stack points, coarse points, and sections

For reference, we spell out the four notions of uniqueness used in the audit.

1. *One geometric coarse point* means that the reduced special fiber of a coarse finite scheme is a singleton. It ignores nilpotents, tangent directions, and all generic information.

2. *One geometric stack lift* means one isomorphism class in a geometric fiber. Its object can still have a nontrivial automorphism group.

3. *A unique scheme-valued section* means that two morphisms $\operatorname{Spec} R \to Y$ over the base which agree on $\operatorname{Spec} K$ are equal. This is a separatedness statement.

4. *A unique stack-valued lift* means uniqueness up to a specified 2-isomorphism, together with uniqueness of that 2-isomorphism. The second part requires trivial relative inertia, which follows here from representability.

These implications do not reverse in general. The algebra $k[T]/(T^\mu)$ has one geometric point but remembers length μ. The classifying stack BG has one geometric isomorphism class over an algebraically closed field but nontrivial automorphisms. A separated finite scheme can have many sections with different generic values. H3 needs the third notion, while its current last paragraph explicitly proves only the first and then invokes categorical coarse descent for the rest.

In the quotient model all four levels can be controlled. The stack map is representable, so relative automorphisms vanish. The coarse map is finite, so a fixed rational generic section extends uniquely. The invariant ring computes its nonreduced special fiber. What remains missing is not a fifth notion of uniqueness, but the global theorem saying that this quotient model is the repository's chosen Hurwitz contraction.

C Why completion does not supply the missing global morphism

Suppose Y and Z are finite over a normal base S, and suppose their completed local rings happen to be isomorphic at corresponding closed points. To conclude $Y \cong Z$ one normally needs a global S-morphism $Y \to Z$ (or compatible formal isomorphisms satisfying effectivity and descent), plus hypotheses such as finiteness and birationality. An unindexed collection of local isomorphisms does not specify how generic components are permuted and need not glue on overlaps.

The same warning applies to conductor squares. For a finite birational extension $A \subset \overline{A}$, the conductor is

$$\mathfrak{c}_A = \{x \in \overline{A} : x\overline{A} \subset A\}.$$

It depends on the embedded subring $A \subset \overline{A}$, not only on $\overline{A}$ and the valuation of an ideal on each branch. Two different subrings of one normalization may have the same conductor exponent while imposing different gluing conditions on residue classes. Therefore the multicluster exponent calculation in H2 is valuable evidence but cannot identify the full normalization–conductor pushout until the contracted downstairs algebra has been identified.

There is a clean route once a morphism exists. If $Y \to Z$ is finite birational and Z is normal, then it is an isomorphism. If both are finite normal S-schemes and a generic isomorphism identifies them with the integral closure of S in the same fixed generic algebra, that generic isomorphism extends uniquely. In both arguments the phrase "same generic algebra" includes an actual identification, and the global morphism or common embedding is essential. This is exactly what item 6 of (CMP) asks H2 to construct.

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